II [CB 213.14.0;2] Customer Block

1 Function Definition

1.1 Architecture specification (static view)

Task

The procedure to reprogramming of ECU Flash memory.

2 Function Description

2.1 Physical basics

This document describes the procedure of reprogramming Flash ECU memory. It makes use of UDS Services and communication is done via CAN-interface.

2.1.1 Prerequisites

2.1.1.1 Required environment conditions

The typical conditions that the ECU needs to be reprogrammed are:

- Ignition on
- vehicle speed = 0
- Engine Speed = 0

for more details please refer to the Drive **DiagnosticSessionControl** module documentaion.

2.1.1.2 Possible combinations of code and data for reprogramming

It is only possible to have the following two combinations of code and data for reprogramming:

1. code block and data block in one complete sequence, or
2. data block only.

It is not recommended to reprogram only the code block of the ECU. It can not be ensured that the existing data block is compatible with the new drive software in the code block. If the two blocks do not match, the ECU will start in reprogramming mode and wait for a reprogramming.

2.1.1.3 Behaviour after previous error in programming

In case of a previous incomplete Flash programming and hence a corrupted memory content, the ECU starts in Boot mode in the next startup and waits for a new programming.

2.1.2 Process Control

Firstly, the ECU has to be Switched into EXTSession,03h using service 10h, **DiagnosticSessionControl** . After this ControlDTCSettings service is requested to switch off the recording of the DTCs in the system. To reduce the busload when reprogramming the ECU, it is recommended to use the service **28h, Communication control - Disable Non Diagnostics messages**. This is used to disable the scheduled sending of CAN messages by other nodes. The use of the above 3 services is optional and Normally these are requested on Functional CAN ID as a broadcast request.

Then ECU has to be switched into programming mode. This is done by requesting *ECUProgrammingMode, 02h* using service **10h, DiagnosticSessionControl**.

If the *ECUProgrammingMode, 02h* is called successfully (positive response), the ECU prepares itself for programming and loads executable code into RAM area.

As a next step, it is necessary to authorize reprogramming to the ECU. This authorisation is done by correctly unlocking the ECU by Seed&Key SecurityAccess procedure.

After successful SecurityAccess and an unlocked ECU, the figerprints(10bytes of tester serial number and 8 bytes of date:DD MM YYYY) are written to ECU using the service **2Eh, WriteDataByIdentifier.**

After this the reprogramming of entire blocks of the ECU memory can be executed. The internal memory layout consists of totally 2 code blocks(0x00,0x01) and 1 data block(0x02). A programming can only be executed in a predefined order.

Reprogramming sequence is always standardized. First the requested area is erased (using service **31h, StartRoutineByLocalIdentifier**), then the data transfer is initiated with **34h, RequestDownload**. The data transfer itself is executed using service **36h, TransferData** and terminated using service **37h, RequestTransferExit** and finally verified by the checkMemory routine (using service **31h, StartRoutineByLocalIdentifier**).

The ECU can then be reset using service **11h, ECUReset**.

Finally, DTC settings, scheduler messages have to be reenabled if they were switched off previously. This is done using Services **10h, DiagnosticSessionControl - ExtendedSession,CommunicationControl - EnableNonDiagnosticsMessages,ControlDTCSettings - DTC ON**. (DTC settings service is not supported in future)

2.1.2.1 programmable memory areas

The below table is an overview of the present memory areas for IFX Device 4+

Block type	Block name	Area Identifier	start address of memory area	size of memory area
Code	CB	0x00	0x80040000	0x00033FFF
	ASW0	0x01	0x80138000	0x00147FFF

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Block type	Block name	Area Identifier	start address of memory area	size of memory area
	ASW1		0x80300000	0x001FFFFF
	ASW2		0x80600000	0x001FFFFF
	ASW3		0x80900000	0x0007FFFF
Data	DS0	0x02	0x80290000	0x0006FFFF
	DS1		0x80510000	0x0006FFFF
	DS2		0x80810000	0x0006FFFF
	DS3		0x809C0000	0x0003FFFF
	VDS		0x80280000	0x0000FFFF
	VDS1		0x80500000	0x0000FFFF
	VDS2		0x80800000	0x0000FFFF
	VDS3		0x80980000	0x0003FFFF

The below table is an overview of the present memory areas for IFX Device 5+ BA step

Block type	Block name	Area Identifier	start address of memory area	size of memory area
Code	CB	0x00	0x80040000	0x00033FFF
	ASW0	0x01	0x80138000	0x00157FFF
	ASW1		0x80300000	0x001FFFFF
	ASW2		0x80600000	0x001FFFFF
	ASW3		0x80900000	0x001FFFFF
	ASW4		0x80C00000	0x001FFFFF
	ASW5		0x80F00000	0x0007FFFF
Data	DS0	0x02	0x802A0000	0x0005FFFF
	DS1		0x80510000	0x0006FFFF
	DS2		0x80810000	0x0006FFFF
	DS3		0x80B10000	0x0006FFFF
	DS4		0x80E10000	0x0006FFFF
	DS5		0x80F90000	0x0004FFFF
	VDS		0x80290000	0x0000FFFF
	VDS1		0x80500000	0x0000FFFF
	VDS2		0x80800000	0x0000FFFF
	VDS3		0x80B00000	0x0000FFFF
	VDS4		0x80E00000	0x0000FFFF
	VDS5		0x80F80000	0x0000FFFF

The below table is an overview of the present memory areas for IF3+ device

Block type	Block name	Area Identifier	start address of memory area	size of memory area
Code	CB	0x00	0x80040000	0x00033FFF
	ASW0	0x01	0x80138000	0x00147FFF
	ASW1		0x80300000	0x001FFFFF
Data	DS0	0x02	0x80290000	0x0006FFFF
	DS1		0x80510000	0x0006FFFF
	VDS		0x80280000	0x0000FFFF
	VDS1		0x80500000	0x0000FFFF

The below table is an overview of the present memory areas for ST3+ Bernina Cut 1 and C 1.2.

Block type	Block name	Area Identifier	start address of memory area	size of memory area
Code	CB	0x00	0x08FD8000	0x00027FFF
	ASW0	0x01	0x09000000	0x001FFFFF
	ASW1		0x09200000	0x0017FFFF

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Block type	Block name	Area Identifier	start address of memory area	size of memory area
Data	DS0	0x02	0x09380000	0x0017FFFF
	VDS		0x09500000	0x000BFFFF

2.1.2.2 Programming of complete ECU software - Code and Data area

The following table represents the complete reprogramming of the ECUs program and data areas. The first column contains the maximum timing parameters, the second gives information about the current programming step. After this, the single UDS services are described with their respective addressing modes. The last column informs about the response from the ECU. The responses are always physical.Below is an example for JDP Device 4 Cut2

time	Sequence	client (Tester)	Request Type	Server (ECU)
	Preparation	DiagnosticSessionControl[ExtendedDiagnosticMode] Request 10 83	functio-nal [3]	
P2				DiagnosticSessionControl[ExtendedDiagnosticMode] Positive Response No response is expected (in case of positive response expected 50 03 xx xx xx xx)
S3		ControlDTCSettings[DTC OFF] Request [2] 85 82	functio-nal [3]	
P2				ControlDTCSettings[DTC OFF]Positive Response No response is expected (in case of positive response expected C5 02)
S3		CommunicationControl[DisableNonDiagnosticsMessages] Request [1] 28 81 01	functio-nal [3]	
P2				CommunicationControl[DisableNonDiagnosticsMessages] Positive Response No response is expected (in case of positive response expected 68 01)
S3 **S3** **S3** **P2**		DiagnosticSessionControl[ECUProgrammingMode] Request 10 02	physical	DiagnosticSessionControl[ECUProgrammingMode] Positive Response 50 02 xx xx xx xx
S3		securityAccess[RequestForSeed] #1 Request 27 01	physical	
P2				securityAccess[sendSeed] #1 Positive Response 67 01 + 8 bytes of seed
S3		securityAccess[SendKey] #2 Request 27 02 + 8 bytes of key	physical	
P2				securityAccess[Unlocked] #2 Positive Response 67 02
S3		writeDataByLocalIdentifier[fingerprint-Tester serial number] Request 2E F1 98+ 10 bytes of data	physical	
P2				writeDataByLocalIdentifier Positive Response 6E F1 98
S3		writeDataByLocalIdentifier[fingerprint - Programing date DD MM YYYY] Request 2E F1 99 + 8 bytes of data	physical	
P2				writeDataByLocalIdentifier Positive Response 6E F1 99

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time	Sequence	client (Tester)	Request Type	Server (ECU)
S3	CB block programming	startRoutineByLocalIdentifier[EraseFlash CodeBlock=0x00] Request 31 01 FF 00 01 00	physical	
P2				startRoutineByLocalIdentifier Positive Response 71 01 FF 00 00
S3		requestDownload[CB Area] Request 34 00 44 08 FD 00 00 00 02 00 00	physical	
P2				requestDownload Positive Response 74 20 0F FF
S3		transferData[CB] #1 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #1 Positive Response 76 + 1 byte of received block sequence counter
S3		transferData[CB] #2 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #2 Positive Response 76 + 1 byte of received block sequence counter
...		...		...
S3		transferData[CB] #n Request 36+1byte of block sequence counter+data	physical	
P2				transferData #n Positive Response 76 + 1 byte of received block sequence counter
S3		requestTransferExit Request 37	physical	
P2				requestTransferExit Positive Response 77
S3		startRoutineByLocalIdentifier[CheckMemory] Request 31 01 02 02 01 00	physical	
P2				startRoutineByLocalIdentifier Positive Response 71 01 02 02 00
S3	ASW0,ASW1 blocks programming	startRoutineByLocalIdentifier[EraseFlash CodeBlock=0x01] Request 31 01 FF 00 01 01	physical	
P2				startRoutineByLocalIdentifier Positive Response 71 01 FF 00 00
S3		requestDownload[ASW0] Request 34 00 44 09 00 00 00 00 30 00 00	physical	
P2				requestDownload Positive Response 74 20 0F FF
S3		transferData[ASW0] #1 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #1 Positive Response 76 + 1 byte of received block sequence counter
S3		transferData[ASW0] #2 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #2 Positive Response 76 + 1 byte of received block sequence counter
...		...		...
S3		transferData[ASW0] #n Request 36+1byte of block sequence counter+data	physical	

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time	Sequence	client (Tester)	Request Type	Server (ECU)
P2				transferData #n Positive Response 76 + 1 byte of received block sequence counter
S3 **P2**		requestTransferExit Request 37	physical	 requestTransferExit Positive Response 77
S3 **P2**		requestDownload[ASW1] Request 34 00 44 09 30 00 00 00 20 00 00	physical	 requestDownload Positive Response 74 20 0F FF
S3 **P2**		transferData[ASW1] #1 Request 36+1byte of block sequence counter+data	physical	 transferData #1 Positive Response 76 + 1 byte of received block sequence counter
S3 **P2** ... **S3** **P2**		transferData[ASW1] #2 Request 36+1byte of block sequence counter+data ... transferData[ASW1] #n Request 36+1byte of block sequence counter+data	physical physical	 transferData #2 Positive Response 76 + 1 byte of received block sequence counter ... transferData #n Positive Response 76 + 1 byte of received block sequence counter
S3 **P2**		requestTransferExit Request 37	physical	 requestTransferExit Positive Response 77
S3 **P2**		startRoutineByLocalIdentifier[CheckMemory] Request 31 01 02 02 01 01	physical	 startRoutineByLocalIdentifier Positive Response 71 01 02 02 00
S3 **P2**	DS0, VDS block programming	startRoutineByLocalIdentifier[EraseFlash DataBlock=0x02] Request 31 01 FF 00 01 02	physical	 startRoutineByLocalIdentifier Positive Response 71 01 FF 00 00
S3 **P2**		requestDownload[DS0] Request 34 00 44 09 58 00 00 00 20 00 00	physical	 requestDownload Positive Response 74 20 0F FF
S3 **P2** **S3** **P2** ... **S3**		transferData[DS0] #1 Request 36+1byte of block sequence counter+data transferData[DS0] #2 Request 36+1byte of block sequence counter+data ... transferData[DS0] #n Request	physical physical physical	 transferData #1 Positive Response 76 + 1 byte of received block sequence counter transferData #2 Positive Response 76 + 1 byte of received block sequence counter ...

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time	Sequence	client (Tester)	Request Type	Server (ECU)
		36+1byte of block sequence counter+data		
P2				transferData #n Positive Response 76 + 1 byte of received block sequence counter
S3		requestTransferExit Request 37	physical	
P2				requestTransferExit Positive Response 77
S3		requestDownload[VDS] Request 34 00 44 09 50 00 00 00 08 00 00	physical	
P2				requestDownload Positive Response 74 20 0F FF
S3		transferData[VDS] #1 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #1 Positive Response 76 + 1 byte of received block sequence counter
S3		transferData[VDS] #2 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #2 Positive Response 76 + 1 byte of received block sequence counter
...		...		...
S3		transferData[VDS] #n Request 36+1byte of block sequence counter+data	physical	
P2				transferData #n Positive Response 76 + 1 byte of received block sequence counter
S3		requestTransferExit Request 37	physical	
P2				requestTransferExit Positive Response 77
S3		startRoutineByLocalIdentifier[CheckMemory] Request 31 01 02 02 01 02	physical	
P2				startRoutineByLocalIdentifier Positive Response 71 01 02 02 00
S3	Restart	ECUReset[powerOnReset] Request	physical	
P2				ECUReset Positive Response
		Wait for ECU initalization(min 500 ms)		
S3		DiagnosticSessionControl[ExtendedDiagnosticMode] Request 10 83	functional [3]	
P2				DiagnosticSessionControl[ExtendedDiagnosticMode] Positive Response No response is expected (in case of positive response expected 50 03 xx xx xx xx)
S3		CommunicationControl[EnableNonDiagnosticsMessages] Request [1] 28 80 01	functional [3]	
P2				CommunicationControl[EnableNonDiagnosticsMessages] Positive Response No response is expected (in case of positive response expected 68 00)

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time	Sequence	client (Tester)	Request Type	Server (ECU)
S3		ControlDTCSettings[DTC ON] Request[2)] 85 81	functio-nal[3)]	
P2				ControlDTCSettings[DTC ON]Positive Response No response is expected (in case of positive response expected C5 01

Hint

[2)] The services **28h, CommunicationControl** is sent with functional request in order to signalize CAN devices other than the ECU to stop non diagnostic communication during ECU reprogramming.

[1)] Service **85h, ControlDTCSettings** is sent with functionalol request in oreder to stop or start the storing of internal errors in the .

[3)] Request send with functional requests are usually suppressed with positive response.This can be customer specific if they want to get the timing parameters from the positive response. Eg. programming request 10 02.

2.1.2.3 Programming of calibration data / Data area only

The following table represents the reprogramming of the ECUs data block only. The first column contains the maximum timing parameters, the second gives information about the current programming step. After this, the UDS services are described with their respective addressing modes. The last column informs about the response from the ECU. The responses are always physical. Below is an example for JDP Device 4 Cut2.

time	Sequence	client (Tester)	Request Type	Server (ECU)
	Preparation	DiagnosticSessionControl[ExtendedDiagnosticMode] Request 10 83	functio-nal[3)]	
P2				DiagnosticSessionControl[ExtendedDiagnosticMode] Positive Response No response is expected (in case of positive response expected 50 03 xx xx xx xx)
S3		ControlDTCSettings[DTC OFF][1)] 85 82	functio-nal[3)]	
P2				ControlDTCSettings[DTC OFF]Positive Response No response is expected (in case of positive response expected C5 02)
S3		CommunicationControl[DisableNonDiagnosticsMessages] Request[2)] 28 81 01	functio-nal[3)]	
P2				CommunicationControl[DisableNonDiagnosticsMessages] Positive Response No response is expected (in case of positive response expected 68 01)
S3		DiagnosticSessionControl[ECUProgrammingMode] Request 10 02	physical	
P2				DiagnosticSessionControl[ECUProgrammingMode] Positive Response 50 02 xx xx xx xx
S3		securityAccess[RequestForSeed] #1 Request 27 01	physical	
P2				securityAccess[sendSeed] #1 Positive Response 67 01 + 8 bytes of seed
S3		securityAccess[SendKey] #2 Request 27 02 + 8 bytes of key	physical	
P2				securityAccess[Unlocked] #2 Positive Response 67 02

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time	Sequence	client (Tester)	Request Type	Server (ECU)
S3		writeDataByLocalIdentifier[fingerprint - Tester serial number] Request 2E F1 98+ 10 bytes of data	physical	
P2				writeDataByLocalIdentifier Positive Response 6E F1 98
S3		writeDataByLocalIdentifier[fingerprint - Programing date DD MM YYYY] Request 2E F1 99 + 8 bytes of data	physical	
P2				writeDataByLocalIdentifier Positive Response 6E F1 99
S3	DS0, VDS block programming	startRoutineByLocalIdentifier[EraseFlash DataBlock=0x02] Request 31 01 FF 00 01 02	physical	
P2				startRoutineByLocalIdentifier Positive Response 71 01 FF 00 00
S3		requestDownload[DS0] Request 34 00 44 09 58 00 00 00 20 00 00	physical	
P2				requestDownload Positive Response 74 20 0F FF
S3		transferData[DS0] #1 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #1 Positive Response 76 + 1 byte of received block sequence counter
S3		transferData[DS0] #2 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #2 Positive Response 76 + 1 byte of received block sequence counter
...		...		...
S3		transferData[DS0] #n Request 36+1byte of block sequence counter+data	physical	
P2				transferData #n Positive Response 76 + 1 byte of received block sequence counter
S3		requestTransferExit Request 37	physical	
P2				requestTransferExit Positive Response 77
S3		requestDownload[VDS] Request 34 00 44 09 50 00 00 00 08 00 00	physical	
P2				requestDownload Positive Response 74 20 0F FF
S3		transferData[VDS] #1 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #1 Positive Response 76 + 1 byte of received block sequence counter
S3		transferData[VDS] #2 Request 36+1byte of block sequence counter+data	physical	
P2				transferData #2 Positive Response 76 + 1 byte of received block sequence counter
...		...		...
S3		transferData[VDS] #n Request	physical	

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time	Sequence	client (Tester)	Request Type	Server (ECU)
		36+1byte of block sequence counter+data		
P2				transferData #n Positive Response 76 + 1 byte of received block sequence counter
S3		requestTransferExit Request 37	physical	
P2				requestTransferExit Positive Response 77
S3		startRoutineByLocalIdentifier[CheckMemory] Request 31 01 02 02 01 02	physical	
P2				startRoutineByLocalIdentifier Positive Response 71 01 02 02 00
S3	Restart	ECUReset[powerOnReset] Request	physical	
P2				ECUReset Positive Response
		Wait for ECU initalization(min 500 ms)		
S3		DiagnosticSessionControl[ExtendedDiagnosticMode] Request 10 83	functio-nal	
P2				DiagnosticSessionControl[ExtendedDiagnosticMode] Positive Response No response is expected (in case of positive response expected 50 03 xx xx xx xx)
S3		CommunicationControl[EnableNonDiagnosticsMessa-ges] Request[1)] 28 80 01	functio-nal[3)]	
P2				CommunicationControl[EnableNonDiagnosticsMessa-ges] Positive Response No response is expected (in case of positive response expected 68 00)
S3		ControlDTCSettings[DTC ON] Request[2)] 85 01	functio-nal[3)]	
P2				ControlDTCSettings[DTC ON]Positive Response No response is expected (in case of positive response expected C5 01)

Hint

[1)] The services **28h, CommunicationControl** is sent with functional request in order to signalize CAN devices other than the ECU to stop non diagnostic communication during ECU reprogramming.

[2)] Service **85h, ControlDTCSettings** is sent with functional request in order to stop or start the storing of internal errors in the .

[3)] Request send with functional requests are usually suppressed with positive response.This can be customer specific if they want to get the timing parameters from the positive response. Eg. programming request 10 02.

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